

State-robust portfolio selection with kernel-weighted conditional CVaR: a semi-infinite model and grid-restricted exchange algorithm

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Abstract

We study a state-robust portfolio selection problem in which observable financial covariates index a family of kernel-weighted empirical return distributions. The state vector consists of implied volatility and trailing equity-market drawdown. For each state, a Nadaraya–Watson weighting scheme defines a conditional empirical distribution, and portfolio risk is measured by conditional value-at-risk (CVaR). The theoretical model minimizes the largest conditional CVaR over a compact continuous state domain, resulting in a convex semi-infinite program.

We establish continuity, convexity, and existence of an optimal solution. For computation, we distinguish the continuous model from its finite-grid approximation. The implemented algorithm is a constraint-generation method over a prescribed state grid: it alternates between a finite linear programming master problem and exhaustive separation over the grid. We prove finite termination for the grid-restricted problem and provide a continuous-domain error bound when conditional CVaR is uniformly Lipschitz in the state variables.

The empirical analysis compares the proposed method with equal weighting, minimum variance, unconditional CVaR, and a finite-regime CVaR benchmark in a strictly chronological rolling-window design. All tables and figures are generated directly from archived machine-readable outputs. The empirical purpose is to evaluate whether continuous state indexing changes allocations and realized risk, not to presume that state robustness must dominate simpler benchmarks out of sample.

Keywords: portfolio optimization; conditional value-at-risk; side information; semi-infinite programming; kernel weighting; constraint generation.

JEL classification: C44; C61; G11.

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1 Introduction

Mean–variance portfolio optimization is sensitive to estimation errors, especially errors in expected returns [Markowitz, 1952, Goldfarb and Iyengar, 2003]. Robust and distributionally robust optimization address this problem by protecting decisions against parameter or distributional uncertainty [Delage and Ye, 2010, Mohajerin Esfahani and Kuhn, 2018]. These formulations are often unconditional, although return distributions and dependence structures vary with observable market conditions.

Conditional stochastic optimization incorporates side information into data-driven decisions [Bertsimas and Kallus, 2020, Esteban-Pérez and Morales, 2022, Nguyen et al., 2025]. A common alternative in finance is to divide observations into a small number of regimes [Ang and Bekaert, 2002]. Such partitions are interpretable but may introduce artificial discontinuities at regime boundaries.

This paper studies an alternative construction. Observable market states index a family of kernel-weighted empirical distributions. Instead of optimizing only for the currently observed state, we seek one portfolio that controls conditional CVaR across an entire state domain. We therefore call the model *state robust*. It is not a distributionally robust model: conditional on a state, the empirical distribution is fixed by the selected kernel and bandwidth.

The theoretical formulation is a convex semi-infinite program because the constraint index set is continuous. The computational implementation, however, uses a finite state grid. This distinction is important. Exhaustive evaluation of a finite grid does not constitute certified continuous global optimization unless a separate discretization-error bound is established.

Our contributions are:

- (i) a kernel-weighted state-robust CVaR formulation with a compact portfolio set and compact state domain;
- (ii) continuity, convexity, and existence results for the continuous problem;
- (iii) a constraint-generation algorithm that terminates finitely for a fixed finite state grid;
- (iv) a uniform approximation bound connecting the grid problem to the continuous model under an explicit Lipschitz assumption; and
- (v) a chronological empirical protocol comparing the method with transparent portfolio benchmarks.

The term *adaptive* refers exclusively to adaptive generation of active state constraints. It does not mean that the final portfolio is an explicit decision rule depending on the contemporaneous state.

2 Related literature

The paper intersects robust portfolio optimization, conditional stochastic optimization, and semi-infinite programming.

Robust portfolio models protect allocations against parameter uncertainty [Goldfarb and Iyengar, 2003, Fabozzi et al., 2007]. CVaR provides a coherent and computationally tractable tail-risk measure [Rockafellar and Uryasev, 2000, 2002]. Distributionally robust models extend this perspective to ambiguity sets of probability distributions [Delage and Ye, 2010, Mohajerin Esfahani and Kuhn, 2018].

Side-information methods construct decisions using conditional empirical distributions or predictive weights [Bertsimas and Kallus, 2020, Scaillet, 2004]. Recent conditional-DRO formulations additionally protect the conditional distribution against misspecification [Esteban-Pérez and Morales, 2022, Nguyen et al., 2025]. Our formulation differs because it fixes one kernel-weighted empirical distribution for each state and protects against variation over the state index itself.

Semi-infinite programs contain finitely many decision variables and infinitely many constraints [Hettich and Kortanek, 1993, López and Still, 2007]. Exchange and cutting-plane algorithms iteratively add violated constraints. Exact convergence statements depend critically on the quality of the separation oracle. Inexact-oracle results require explicit error assumptions and cannot be inferred solely from the use of a numerical grid [Oustry and Cerulli, 2025].

3 State-dependent empirical distributions

3.1 Returns and lagged states

Let

$$x_t = (x_{1t}, \dots, x_{Nt})^\top \in \mathbb{R}^N$$

denote the asset-return vector from trading day $t - 1$ to day t . Let

$$y_{t-1} = (\log V_{t-1}, D_{t-1})^\top \in \mathbb{R}^2$$

be the information available before x_t is realized, where V_t denotes VIX.

We define positive trailing drawdown by

$$D_t = 1 - \frac{P_t}{\max_{s \in \{\max(1, t-L), \dots, t\}} P_s}, \quad L = 63, \quad (1)$$

so that $D_t \in [0, 1]$. The market index P_t , its source, and its construction must be fixed before the backtest and documented in the replication files.

The lagged pairing (y_{t-1}, x_t) prevents the use of information observed after the return interval begins.

3.2 State domain

For a given training sample, define

$$\mathcal{U} = [\underline{v}, \bar{v}] \times [0, \bar{d}], \quad (2)$$

where

$$\underline{v} = v_{\min} - \delta_v, \quad \bar{v} = v_{\max} + \delta_v,$$

and

$$\bar{d} = \min\{1, d_{\max} + \delta_d\}.$$

The bounds must be calculated using the current training window only. Using full-sample extrema would introduce look-ahead bias.

A rectangular domain may include weakly supported state combinations. Accordingly, we report $\text{ESS}(\theta)$ and conduct sensitivity analysis with a supported domain

$$\mathcal{U}_E = \{\theta \in \mathcal{U} : \text{ESS}(\theta) \geq E_{\min}\}. \quad (3)$$

3.3 Kernel weights

Let $H \in \mathbb{R}^{2 \times 2}$ be symmetric positive definite. For $\theta \in \mathcal{U}$, define

$$p_t(\theta) = \frac{\exp\left[-\frac{1}{2}(y_{t-1} - \theta)^\top H^{-1}(y_{t-1} - \theta)\right]}{\sum_{s=1}^T \exp\left[-\frac{1}{2}(y_{s-1} - \theta)^\top H^{-1}(y_{s-1} - \theta)\right]}. \quad (4)$$

The common normalizing factor of the Gaussian kernel cancels from the ratio. For numerical stability, the implementation evaluates (4) using a log-sum-exp calculation.

The baseline bandwidth is

$$H = T^{-2/(M+4)} \hat{\Sigma}_y = T^{-1/3} \hat{\Sigma}_y, \quad M = 2. \quad (5)$$

Bandwidth sensitivity is evaluated through

$$H_\gamma = \gamma H, \quad \gamma \in \{0.5, 0.75, 1, 1.5, 2\}.$$

The effective sample size is

$$\text{ESS}(\theta) = \frac{1}{\sum_{t=1}^T p_t(\theta)^2}. \quad (6)$$

Because CVaR uses a lower-tail probability τ , we also report

$$\text{ESS}_{\text{tail}}(\theta) = \tau \text{ESS}(\theta).$$

A strictly positive Gaussian weight does not imply adequate local support.

4 State-robust CVaR model

4.1 Feasible portfolios

Let $\bar{w} \in (0, 1]$ denote the maximum asset weight. Define

$$\mathcal{W} = \left\{ w \in \mathbb{R}^N : \mathbf{1}^\top w = 1, 0 \leq w_i \leq \bar{w}, \hat{\mu}^\top w \geq \mu_{\text{target}} \right\}. \quad (7)$$

The condition $N\bar{w} \geq 1$ is necessary for the capped simplex to be nonempty. Let

$$\hat{\mu}_{(1)} \geq \hat{\mu}_{(2)} \geq \cdots \geq \hat{\mu}_{(N)}$$

be the ordered estimated asset means, and define

$$q = \left\lfloor \frac{1}{\bar{w}} \right\rfloor, \quad r = 1 - q\bar{w}.$$

When $r > 0$,

$$\mu_{\max} = \bar{w} \sum_{i=1}^q \hat{\mu}_{(i)} + r\hat{\mu}_{(q+1)}. \quad (8)$$

When $r = 0$, the residual term is omitted. Feasibility requires $\mu_{\text{target}} \leq \mu_{\max}$.

4.2 Conditional CVaR

For portfolio w , define losses

$$\ell_t(w) = -x_t^\top w.$$

At confidence level $\alpha = 1 - \tau$, conditional CVaR is

$$\Phi_\tau(w, \theta) = \min_{z \in \mathbb{R}} \left\{ z + \frac{1}{\tau} \sum_{t=1}^T p_t(\theta) [\ell_t(w) - z]_+ \right\}. \quad (9)$$

Any minimizer satisfies

$$z^*(w, \theta) \in \text{VaR}_{1-\tau}(\ell(w) \mid \theta). \quad (10)$$

For a discrete weighted distribution, the VaR minimizer need not be unique, although the optimized CVaR value is well defined.

4.3 Continuous state-robust problem

Define

$$G(w) = \max_{\theta \in \mathcal{U}} \Phi_\tau(w, \theta). \quad (11)$$

The theoretical continuous model is

$$v^* = \min_{w \in \mathcal{W}} G(w). \quad (12)$$

Equivalently,

$$\min_{w, \eta} \quad \eta \quad (13)$$

$$\text{s.t.} \quad \Phi_\tau(w, \theta) \leq \eta, \quad \forall \theta \in \mathcal{U}, \quad (14)$$

$$w \in \mathcal{W}. \quad (15)$$

Proposition 1 (Well-posedness). *Suppose that H is symmetric positive definite, $\mathcal{U}$ is nonempty and compact, and $\mathcal{W}$ is nonempty. Then:*

- (i) $p_t(\theta)$ is positive and continuously differentiable on $\mathcal{U}$;
- (ii) Φ_τ is jointly continuous on $\mathcal{W} \times \mathcal{U}$;
- (iii) $\Phi_\tau(\cdot, \theta)$ is convex on $\mathcal{W}$;
- (iv) G is continuous and convex on $\mathcal{W}$; and
- (v) problem (12) admits an optimal solution.

Proof. The numerator and denominator in (4) are smooth and the denominator is strictly positive. Hence p_t is smooth.

Because $\mathcal{W}$ is compact, portfolio losses are uniformly bounded. The minimization over z in (9) can therefore be restricted to a common compact interval containing all attainable losses. Continuity follows from Berge's maximum theorem. Convexity in w follows from partial minimization of a jointly convex function in (w, z) .

The maximum of a jointly continuous function over compact $\mathcal{U}$ is continuous in w . It is convex because it is a pointwise maximum of convex functions. Existence follows from continuity of G and compactness of $\mathcal{W}$. $\square$

5 Grid-restricted exchange algorithm

5.1 Finite approximation

Let

$$\hat{\mathcal{U}}_K = \{\theta^1, \dots, \theta^K\} \subset \mathcal{U}$$

be a finite grid. The implemented optimization problem is

$$\hat{v}_K = \min_{w \in \mathcal{W}} \hat{G}_K(w), \quad \hat{G}_K(w) = \max_{\theta \in \hat{\mathcal{U}}_K} \Phi_\tau(w, \theta). \quad (16)$$

Problem (16) is a finite robust linear program after introducing state-specific VaR and excess-loss variables.

5.2 Master problem

At iteration k , let

$$\mathcal{U}_k \subseteq \hat{\mathcal{U}}_K$$

be the active state set. The master problem is

$$\text{LB}_k = \min_{w, \eta, z, u} \eta \quad (17)$$

$$\text{s.t.} \quad z_\theta + \frac{1}{\tau} \sum_{t=1}^T p_t(\theta) u_{t\theta} \leq \eta, \quad \forall \theta \in \mathcal{U}_k, \quad (18)$$

$$u_{t\theta} \geq -x_t^\top w - z_\theta, \quad t = 1, \dots, T, \theta \in \mathcal{U}_k, \quad (19)$$

$$u_{t\theta} \geq 0, \quad t = 1, \dots, T, \theta \in \mathcal{U}_k, \quad (20)$$

$$w \in \mathcal{W}. \quad (21)$$

For the resulting portfolio w_k , exhaustive grid separation computes

$$\hat{G}_k = \max_{\theta \in \hat{\mathcal{U}}_K} \Phi_\tau(w_k, \theta) \quad (22)$$

and selects a maximizer $\hat{\theta}_k$.

The stopping residual is

$$r_k = \hat{G}_k - \text{LB}_k. \quad (23)$$

It is a residual for the finite-grid problem, not automatically a certified optimality gap for the continuous problem.

Proposition 2 (Finite termination on a fixed grid). *Suppose every master problem is solved globally and every state in $\hat{\mathcal{U}}_K$ is evaluated exactly during separation. If a newly violated state is added whenever $r_k > \epsilon$, then the algorithm terminates after at most K state additions. At termination,*

$$0 \leq \hat{G}_K(w_k) - \hat{v}_K \leq \epsilon.$$

Proof. Each nonterminating iteration adds a state not already represented in the master problem. Because $\hat{\mathcal{U}}_K$ contains K states, at most K distinct additions are possible. Moreover,

$$\text{LB}_k \leq \hat{v}_K \leq \hat{G}_K(w_k).$$

Thus, when $\hat{G}_K(w_k) - \text{LB}_k \leq \epsilon$,

$$\hat{G}_K(w_k) - \hat{v}_K \leq \hat{G}_K(w_k) - \text{LB}_k \leq \epsilon.$$

□

5.3 Continuous-domain approximation

Define the grid dispersion radius

$$\rho_K = \max_{\theta \in \mathcal{U}} \min_{\hat{\theta} \in \hat{\mathcal{U}}_K} \|\theta - \hat{\theta}\|. \quad (24)$$

Proposition 3 (Grid approximation bound). *Suppose that, uniformly for $w \in \mathcal{W}$,*

$$|\Phi_\tau(w, \theta) - \Phi_\tau(w, \theta')| \leq L_\Phi \|\theta - \theta'\|$$

for all $\theta, \theta' \in \mathcal{U}$. Then

$$\widehat{G}_K(w) \leq G(w) \leq \widehat{G}_K(w) + L_\Phi \rho_K \quad (25)$$

for every $w \in \mathcal{W}$, and

$$\widehat{v}_K \leq v^* \leq \widehat{v}_K + L_\Phi \rho_K. \quad (26)$$

If the exchange algorithm terminates with residual at most ϵ , then

$$G(w_k) - v^* \leq \epsilon + L_\Phi \rho_K. \quad (27)$$

Proof. For each $\theta \in \mathcal{U}$, choose a nearest grid point $\widehat{\theta} \in \widehat{\mathcal{U}}_K$. Lipschitz continuity gives

$$\Phi_\tau(w, \theta) \leq \Phi_\tau(w, \widehat{\theta}) + L_\Phi \rho_K \leq \widehat{G}_K(w) + L_\Phi \rho_K.$$

Taking the maximum over θ establishes (25). Minimizing over w gives (26). Finally,

$$G(w_k) \leq \widehat{G}_K(w_k) + L_\Phi \rho_K \leq \widehat{v}_K + \epsilon + L_\Phi \rho_K \leq v^* + \epsilon + L_\Phi \rho_K.$$

□

Remark 1. A numerical value of $L_\Phi \rho_K$ is informative only if its units, coordinate system, norm, and magnitude relative to portfolio CVaR are reported. A very large bound is mathematically valid but practically vacuous.

6 Empirical protocol

6.1 Data

The empirical universe consists of the Kenneth French 30 value-weighted industry portfolios. VIX is obtained from Cboe. The broad-market series used to construct (1) must be identified exactly.

The final manuscript must report:

- exact source URLs and access dates;
- whether the market series is CRSP, a French market factor, or an industry-portfolio proxy;
- whether any VIX–VXO splice is used and, if so, its exact construction;
- the first and final aligned dates;
- the number of observations; and
- a checksum for the frozen input dataset.

No observation after a rebalancing date may be used to determine state bounds, bandwidths, target returns, or portfolio weights.

6.2 Rolling design

For rebalancing date q , use a trailing training window of 1,260 trading days. The portfolio is evaluated over the following nonoverlapping holding period.

A machine-readable backtest calendar must contain:

Window	Train start	Train end	Hold start	Hold end	Observations
Generated by the backtest script					

Fixed 21-trading-day blocks must be described as 21-day holding periods, not as exact calendar months.

6.3 Benchmarks

The benchmarks are:

1. monthly rebalanced $1/N$;
2. capped minimum variance;
3. capped unconditional empirical CVaR; and
4. capped four-regime CVaR using training-sample median splits.

All optimized models use the same feasible set $\mathcal{W}$. Solver failure must not silently remove the target-return constraint. Every optimization exports its termination status and feasibility residual.

6.4 Turnover and transaction costs

Let g_{iq} be asset i 's gross growth during holding period q . The weights immediately before rebalance $q + 1$ are

$$w_{i,q+1}^- = \frac{w_{i,q}^+ g_{iq}}{\sum_{j=1}^N w_{j,q}^+ g_{jq}}. \quad (28)$$

Turnover is

$$\text{TO}_{q+1} = \frac{1}{2} \sum_{i=1}^N |w_{i,q+1}^+ - w_{i,q+1}^-|. \quad (29)$$

The drift must use the period ending at the current rebalancing date, not the future holding-period return.

For proportional one-way cost c ,

$$r_{q+1}^{\text{net}} = r_{q+1}^{\text{gross}} - c \text{TO}_{q+1}. \quad (30)$$

6.5 Inference

Sharpe-ratio comparisons use paired circular moving-block resampling of the two strategy return series. Unless an inner standard-error estimator is computed in every replication, the procedure must not be called studentized.

The manuscript reports:

- the block length;
- number of replications;
- random seed;
- confidence-interval construction;
- definition of the empirical p -value; and
- multiplicity-adjusted results for several benchmarks.

7 Empirical results

All results in this section must be generated directly by the archived analysis pipeline. Numerical values must not be manually entered into plotting scripts.

7.1 Performance

Required generated file: `generated/performance_table.tex`.

This table must be regenerated after correcting the data, state construction, grid definition, turnover timing, and solver-status handling.

The interpretation must distinguish absolute return from downside protection. If the proposed method has lower Sharpe, higher CVaR, or deeper drawdown than a benchmark, this must be stated directly.

7.2 Crisis-period results

Required generated file: `generated/crisis_table.tex`.

Crisis dates must be fixed before inspecting strategy performance. Subperiod drawdown calculations must include initial wealth $W_0 = 1$.

7.3 Grid resolution and computational diagnostics

The minimum required resolutions are

$$21 \times 21, \quad 41 \times 41, \quad 81 \times 81, \quad 161 \times 161.$$

For each resolution, report:

- grid objective value;
- exchange objective value;
- grid residual;
- solve time;
- number of master iterations;

- number of active states;
- portfolio L_1 distance; and
- objective-value difference.

A large portfolio L_1 distance must not be described as negligible merely because objective values are close.

7.4 ESS and bandwidth sensitivity

Report the minimum, 5th percentile, median, mean, and maximum ESS of all active states. Also report τ ESS. Sensitivity analysis must include both bandwidth scaling and minimum-ESS restrictions.

7.5 Statistical inference

Required generated file: `generated/bootstrap_table.tex`.

The table must identify whether intervals are percentile, basic, studentized, or normal-approximation block-bootstrap intervals.

Failure to reject equality must not be interpreted as evidence that strategies are equivalent. It indicates only insufficient evidence of a difference under the selected test.

8 Discussion

The continuous formulation offers an interpretable method for studying portfolio risk over a family of covariate-indexed empirical distributions. Its main limitation is statistical support in extreme state regions. A Gaussian kernel assigns positive mass everywhere, but the effective number of observations may nevertheless be very small.

The grid-restricted exchange algorithm can reduce the size of the master LP relative to a monolithic finite-grid formulation. Its computational advantage does not remove the curse of dimensionality: Cartesian grid size grows exponentially with state dimension, and local kernel support deteriorates in high dimensions.

The model should also not be expected to dominate unconditional CVaR in every crisis. Worst-state conditional CVaR and realized drawdown are different criteria. Empirical conclusions must therefore be based on the corrected out-of-sample evidence rather than on the model's robust interpretation.

Potential extensions include data-supported state domains, conditional ambiguity sets, turnover regularization, and certified global separation for low-dimensional state spaces.

9 Conclusion

We formulated a state-robust portfolio problem that minimizes the maximum kernel-weighted conditional CVaR over a compact market-state domain. The continuous model is a convex semi-infinite program and admits an optimal solution under standard compactness and bandwidth assumptions.

The practical implementation is a grid-restricted constraint-generation algorithm. It terminates finitely for a fixed grid and yields an ϵ -optimal solution to the finite-grid problem. Approximation of the continuous model additionally depends on the grid dispersion and a valid uniform Lipschitz bound.

The empirical contribution must be evaluated after the complete pipeline has been rerun with synchronized data, code, tables, and figures. In particular, claims of superior downside protection or continuous global optimality should be made only if supported by the regenerated evidence and a quantitatively useful approximation certificate.

Data and code availability

The replication archive is available at:

<https://github.com/MadBezoui/Robust-SIP-Portfolio>

The final submitted version should identify a permanent release tag and commit hash. The archive must include:

- source-data documentation and checksums;
- the exact backtest calendar;
- solver statuses for every window;
- machine-readable portfolio weights and returns;
- raw bootstrap replications;
- convergence histories;
- a Julia `Manifest.toml`;
- pinned Python dependencies; and
- scripts that generate every table and figure without manual values.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Use of generative artificial intelligence

If generative artificial intelligence was used for language editing, software assistance, or manuscript preparation, its role should be disclosed according to the target journal’s policy. The authors remain responsible for all mathematical statements, references, code, data, and empirical results.

References

- Andrew Ang and Geert Bekaert. International asset allocation with regime shifts. *The Review of Financial Studies*, 15(4):1137–1187, 2002. doi: 10.1093/rfs/15.4.1137.
- Dimitris Bertsimas and Nathan Kallus. From predictive to prescriptive analytics. *Management Science*, 66(3):1025–1044, 2020. doi: 10.1287/mnsc.2018.3253.
- Erick Delage and Yinyu Ye. Distributionally robust optimization under moment uncertainty with application to data-driven problems. *Operations Research*, 58(3):595–612, 2010. doi: 10.1287/opre.1090.0741.
- Adrián Esteban-Pérez and Juan M. Morales. Distributionally robust stochastic programs with side information based on trimmings. *Mathematical Programming*, 195(1–2):1069–1105, 2022. doi: 10.1007/s10107-021-01683-1.
- Frank J. Fabozzi, Dashan Huang, and Guofu Zhou. Robust portfolio optimization. *The Journal of Portfolio Management*, 33(3):40–48, 2007. doi: 10.3905/jpm.2007.684751.
- Donald Goldfarb and Garud Iyengar. Robust portfolio selection problems. *Mathematics of Operations Research*, 28(1):1–38, 2003. doi: 10.1287/moor.28.1.1.14258.
- Rainer Hettich and Kenneth O. Kortanek. Semi-infinite programming: Theory, methods, and applications. *SIAM Review*, 35(3):380–429, 1993. doi: 10.1137/1035089.
- Marco A. López and Georg Still. Semi-infinite programming. *European Journal of Operational Research*, 180(2):491–518, 2007. doi: 10.1016/j.ejor.2006.02.040.
- Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952. doi: 10.1111/j.1540-6261.1952.tb01525.x.
- Peyman Mohajerin Esfahani and Daniel Kuhn. Data-driven distributionally robust optimization using the Wasserstein metric: Performance guarantees and tractable reformulations. *Mathematical Programming*, 171(1–2):115–166, 2018. doi: 10.1007/s10107-017-1172-1.
- Viet Anh Nguyen, Fan Zhang, Shanshan Wang, José Blanchet, Erick Delage, and Yinyu Ye. Robustifying conditional portfolio decisions via optimal transport. *Operations Research*, 73(5):2801–2829, 2025. doi: 10.1287/opre.2021.0243.
- Antoine Oustry and Martina Cerulli. Convex semi-infinite programming algorithms with inexact separation oracles. *Optimization Letters*, 19(3):437–462, 2025. doi: 10.1007/s11590-024-02148-3.
- R. Tyrrell Rockafellar and Stanislav Uryasev. Optimization of conditional value-at-risk. *Journal of Risk*, 2(3):21–42, 2000. doi: 10.21314/JOR.2000.038.
- R. Tyrrell Rockafellar and Stanislav Uryasev. Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7):1443–1471, 2002. doi: 10.1016/S0378-4266(02)00271-6.

Olivier Scaillet. Nonparametric estimation and sensitivity analysis of expected shortfall. *Mathematical Finance*, 14(1):115–129, 2004. doi: 10.1111/j.0960-1627.2004.00184.x.